The Paleomagnetism of the Precambrian Dolerites of Eastern Southern Rhodesia, an Example of Geologic Correlation by Rock Magnetism

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Abstract. Thirty sites have been sampled in dolerites which intrude the Precambrian shield and the Umkondo system of eastern Southern Rhodesia. The natural remanent magnetization is random at sixteen of the twenty-five sites used in the final analysis. In many cases it has been necessary to resort to the use of high alternating fields to successfully remove later components of magnetization, which are thought to be the result of isothermal remanent magnetization due to lightning currents. The directions of magnetization after treatment in alternating fields form two distinct groups about 90° apart, which indicates that these dolerites represent two quite distinct magmatic episodes. These directions are given by $D = 292°$, $I = +51°$ (pole position 7°N, 20°W, $A_{95} = 7½°$), and $D = 182½°$, $I = -10½°$ (pole position 65°N, 42½°E, $A_{95} = 6°$). The former group, which intrudes the Rhodesian Precambrian shield, are termed the *Mashonaland dolerites,* and the latter, which are associated with the Umkondo system, are termed the *Umkondo dolerites.* Paleomagnetic evidence is presented to show that the Mashonaland dolerites have intruded the Umkondo dolerites, which are therefore the older of the two sets of intrusions. Potassium-argon age determinations on some of the samples of Mashonaland dolerite used in this study indicate that this group of intrusions is at least 1600 m.y. old. This would indicate that the Umkondo dolerites and therefore the Umkondo system itself are at least this old.

Introduction. Throughout the eastern part of Southern Rhodesia in an area between longitudes 30° and 33°E and between latitudes 17° and 21°S, a large number of dolerite dikes, sheets, and sills have intruded the Precambrian shield of Southern Rhodesia. The dolerites crop out over an area of 80,000 km² and are seen to intrude the Great Dyke, the basement granites and gneisses, and the Umkondo-Frontier system (Figure 1). In the southeast part of Southern Rhodesia, around Melsetter, the dolerites intruding the Umkondo system take the form of large sills up to 200 m thick which can be traced for many kilometers. The geology of this part of the Umkondo system, along with its intrusions, has been described by *Swift* [1962] and *Watson* [1964].

In the northeast part of Southern Rhodesia, around Inyanga, a further sequence of thick sills up to 450 m thick are associated with what are presumed to be remnants of the Umkondo system in this area [*Tyndale-Biscoe,* 1958]. The sheets and dikes intruding the granites, schists, and gneisses of the older Precambrian basement are mainly demarked by the occurrences of red soil which show up conspicuously in the sandy soil derived from the granite. Most of these intrusions (see Figure 1) have been mapped from aerial photographs, and their considerable extent has only recently been appreciated.

In the Umkondo system in the southeast corner of Southern Rhodesia, one of the dolerite sills is overlain unconformably by the local base of the Karroo system. *Swift* [1961], correlating the sills in the Inyanga district with those in the Melsetter district, believes that, because the north-south extent is comparable with that of the Great Dyke, the two are probably contemporaneous. On the other hand, *Nicolaysen et al.* [1962] think that the dolerites around Shawa [see *Gough and Brock,* 1964] intruding the basement granite are of Karroo age, mainly because of their wide occurrence. Before this investigation it was not known whether these intrusions represented one magmatic episode or several.

The investigation of these intrusions arose

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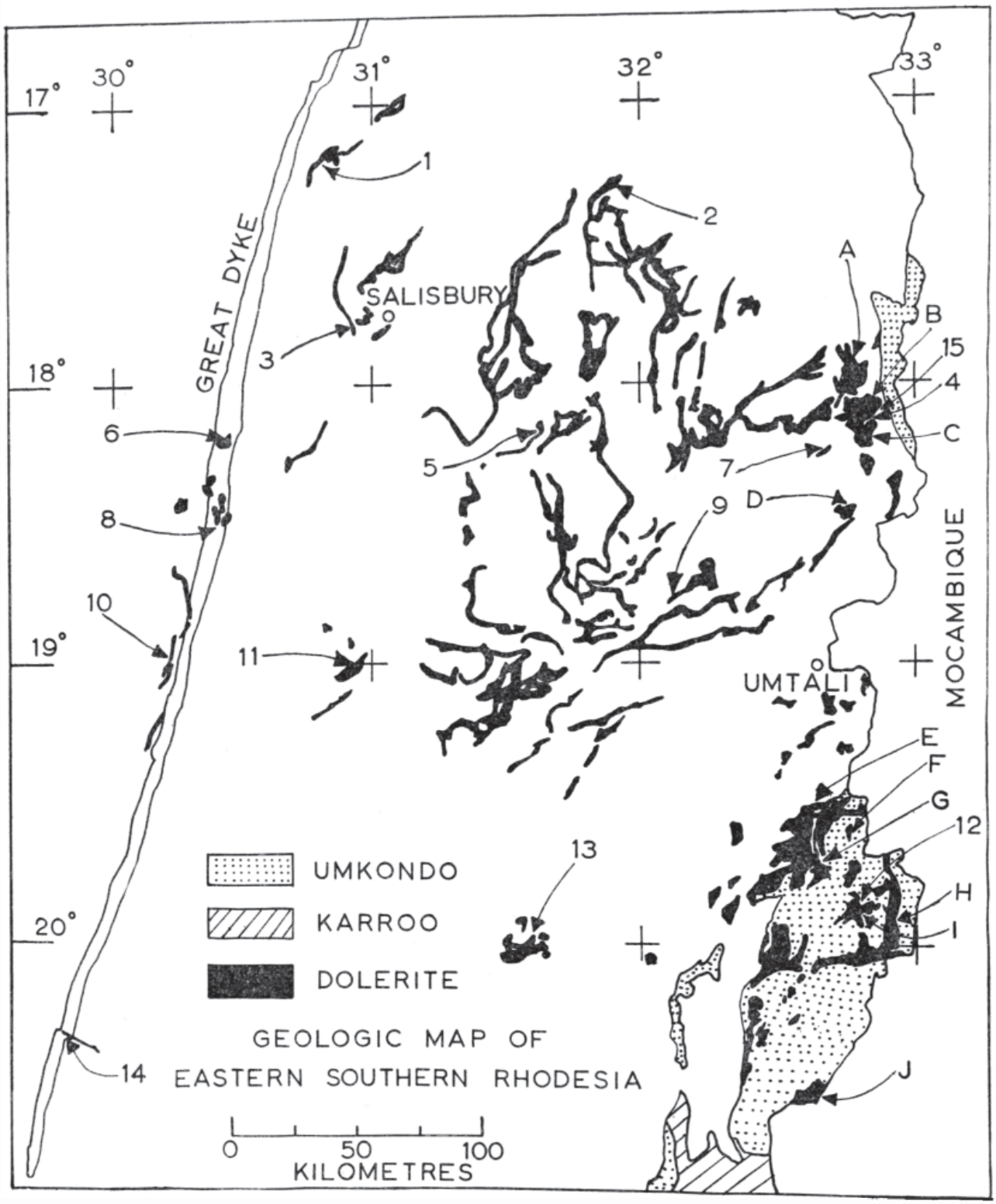


Fig. 1. Geological sketch map of eastern Southern Rhodesia showing sampling sites.

out of a study of the paleomagnetism of the Great Dyke by *McElhinny and Gough* [1963]. In this study one of the sites sampled in the norite at the Umsweswe River was found to have a direction of magnetization quite different from that characteristic of the rest of the Great Dyke. The most likely explanation of this anomalous direction was that it occurred from the reheating of the norite at the time of intrusion of the dolerite sills, dikes, and sheets, remnants of which outcrop in the area. To test this hypothesis we sampled several of these sills,

and we then decided to extend the sampling to include the rest of the dolerites in the eastern half of the country.

Sampling and field methods. Thirty sites, spread throughout the area of occurrence of the dolerites, were sampled. As a result of some preliminary sampling, two areas of interest were later sampled more extensively. These were the Inyanga district and the Melsetter district along the eastern border of Southern Rhodesia.

We used a Graham-Keiller type portable sampling drill [*Graham and Keiller*, 1960], usually drilling eight or ten cores, 15 to 20 cm long, at each site. At one site the cores were extended to a depth of 60 cm, and only four were taken. These cores were oriented relative to the sun's azimuth by the method described by *Gough and Opdyke* [1963]. The positions of the sites are shown in Figure 1 and the geographical coordinates are given in Table 1.

Because of the heavy oxidizing conditions which exist in tropical countries, good outcrops are often hard to find. Also, the high probability of lightning striking the ground in southern Africa makes the selection of good sampling sites difficult. Where possible the sites were located in road cuts, but in an underdeveloped country these are rare. The majority of the samples were taken from natural exposures. Experience has shown that the best types of natural exposure are in river beds, where the rock is reasonably fresh and where the river valley affords some protection from lightning. The individual cores were usually taken at least 2 m apart in order to minimize the effect of a lightning strike [*Graham*, 1961].

The natural remanent magnetization. The directions and intensities of the natural remanent magnetization (NRM) were measured by means of a spinner magnetometer [*Gough*, 1964]. Directions were measured with an uncertainty of 1° and intensities within 5 per cent.

TABLE 1. Site Directions for the Precambrian Dolerites of Southern Rhodesia

Site	Site Position				M_0					M_{ac}					Site Pole	
No.	Lat. S.	Long. E	N	E	D	I	R	k	B_{ac}	D	I	R	k	α_{95}	Lat.	Long.
1	17°10′	30°51′	9	1			2.31*		300	133	−56	8.830	47	7½	20½°N	8°W
2	17°27′	31°58′	8	2			3 09*		800	90½	−47½	7.433	12 4	16½	8°S	30½°W
3	17°50′	30°55′	10	0			2.58*		400	113½	−48½	9.879	75	5½	10½°N	23½°W
4	18°09′	32°51′	10	0			4.01*		800	105	−53½	9.727	33	8½	1½°N	20½°W
5	18°12′	31°42′	8	0			3 98*		800	298½	+44	7.850	47	8	16°N	23½°W
6	18°13′	30°26′	4	0	38	+58	1.85	1.4	800	120	−49	3.986	215	6½	15°N	20½°W
7	18°17′	32°41′	8	2			2.49*		300	263½	+51	7 058	7.5	21½	15°S	28½°W
8	18°30′	30°24′	10	0	108	−21	9.06	9.6	150	114	−33½	9.903	94	4½	15½°N	33½°W
9	18°45′	32°12′	10	0			3 39*		300	291	+63	9.932	133	4	1°N	9½°W
10	19°01′	30°13′	10	0	91½	−47	7.52	3 6	100	116½	−40½	9.907	97	4½	15°N	28°W
11	19°02′	30°55′	8	0			3 20*		400	111	−46½	7.987	539	2½	8½°N	25½°W
12	19°52′	32°48′	8	0	190½	−59½	5 38	2.7	200	132	−54½	7.765	30	10½	18½°N	7°W
13	19°59′	31°35′	10	0			4 79*		400	279	+73½	9 463	17	13	12½°S	0½°E
14	20°22′	29°49′	8	0			3 86*		200	118½	−51	7.706	24	11½	11½°N	20°W
15†	18°07′	32°53′	5	0			3 23*		800	14	+53	4.936	62	5½		
A	18°01′	32°47′	9	1			3.24*		400	186	−17	8 757	33	9	63°N	49½°E
B	18°06′	32°51′	5	0	93	+21	4.72	14	500	171½	−10	4 985	267	4½	65½°N	12°E
C	18°12′	32°51′	8	0			3 44*		800	184½	−8	7.671	21	12½	67½°N	44½°E
D	18°27′	32°46′	10	0	95½	−4	6.41	2 5	1000	168	−5½	9.288	12.6	14	66°N	21°E
E	19°32′	32°38′	10	0	182	−51½	9 40	15	100	185	−3½	9.902	92	5	68°N	46°E
F	19°36′	32°48′	8	0	90½	−65	5 83	6 1	300	179½	−13	7.966	206	4	64°N	31½°E
G	19°47′	32°41′	9	0	223	−52	8 47	15	100	204	−6	8.954	174	4	57°N	81°E
H	19°51′	32°57′	10	0			2.73*		300	185	−2½	9.576	21	10½	68½°N	46½°E
I	19°53′	32°48′	8	0			4 32*		250	176	−14	6 766	5.7	25½	62½°N	24°E
J	20°32′	32°40′	7	1			1.08*		800	180½	−10	6.570	14	16½	64½°N	34°E

M_0 gives the NRM results, and M_{ac} gives those after demagnetization in a peak alternating field of B_{ac} gauss.

N is the number of samples (i.e. cores) included in the analysis and E is the number excluded.

R is the length of the resultant of the N unit vectors and D and I are the declination and inclination of its direction.

α_{95} is the semiangle of the cone of 95 per cent confidence, and $k = (N - 1)/(N - R)$ is the best estimate of Fisher's dispersion parameter.

* The resultant R is less than the value R_0 tabulated by *Watson* [1956] which is exceeded in only 5 per cent of the cases by the resultant of N randomly directed vectors.

† Site 15 is excluded; see text.

The parameters of the NRM at each site are presented in Table 1. The directions at all sites are widely scattered, and at sixteen of the twenty-five sites used in the final analysis the resultant R of the N vectors is less than the value R_0 tabulated by *Watson* [1956], which is exceeded in only 5 per cent of the cases by the resultant of N randomly directed vectors. At all these sites the directions may be considered random according to Watson's criterion.

Alternating field demagnetization. The samples from all sites were subjected to alternating-field demagnetization using the apparatus and techniques outlined by *McElhinny and Gough* [1963]. At those sites where the NRM directions were not random according to Watson's criterion, but were still somewhat scattered, the directions of magnetization generally reached a stable end point in low fields. At some of these sites the NRM directions exhibited streaking toward the present earth's field, a characteristic of viscous magnetization. Figure 2 shows this effect at site E, and this viscous magnetization was removed in a peak field of 100 gauss.

If the NRM directions were randomly distributed, high fields, exceeding 200 gauss, were required to align the directions. Figure 3 shows the results from site 3, where a randomly distributed NRM forms an excellent group when demagnetized in a peak field of 400 gauss. At site D the randomly distributed NRM required a peak field of 1000 gauss (the maximum available on the equipment in use) to obtain the best grouping. Figure 4 shows the directions after successive demagnetizations in peak fields of 300, 600, 800, and 1000 gauss. The parameters for this site at each of these demagnetizing fields are given in Table 2. The value of

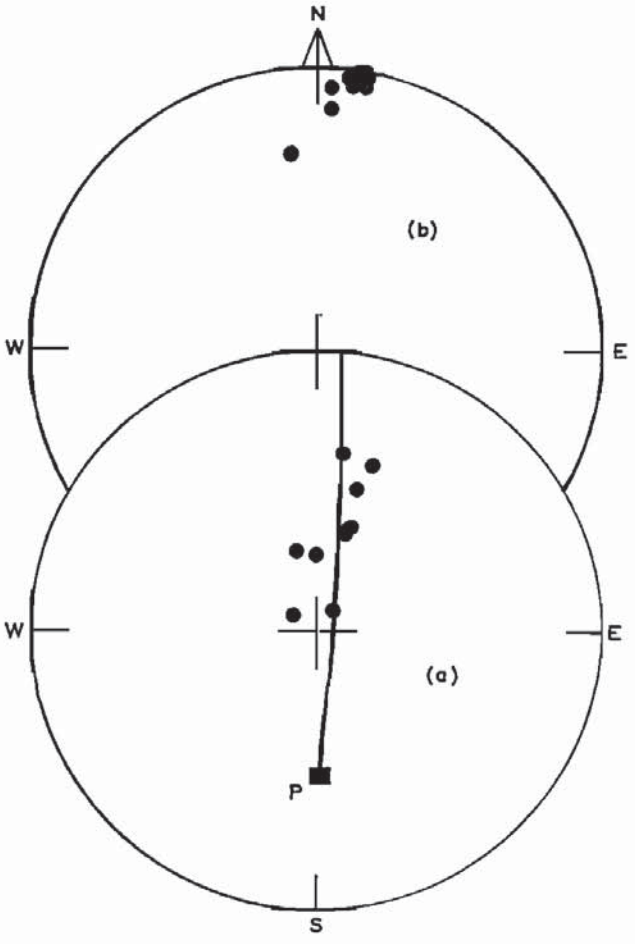


Fig. 2. Directions of magnetization at site E. (*a*) NRM, and (*b*) after demagnetization in a peak alternating field of 100 gauss. Lower-hemisphere stereographic projection is used; all points south-seeking directions down. P is the present axial dipole field direction.

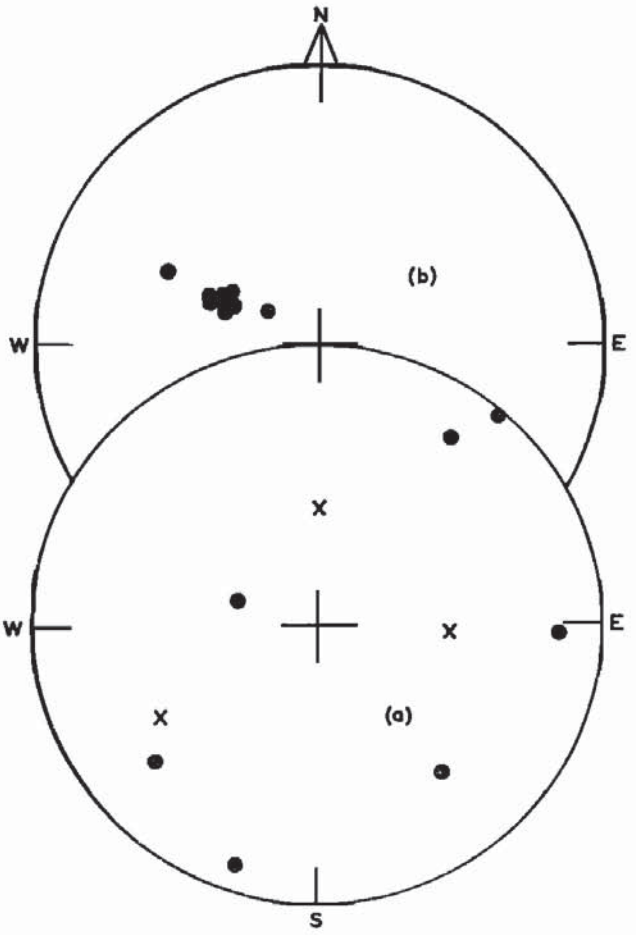


Fig. 3. Directions of magnetization at site 3. (*a*) NRM, and (*b*) after demagnetization in a peak alternating field of 400 gauss. Lower-hemisphere stereographic projection. Solid circles, south-seeking; crosses, north-seeking, direction down.

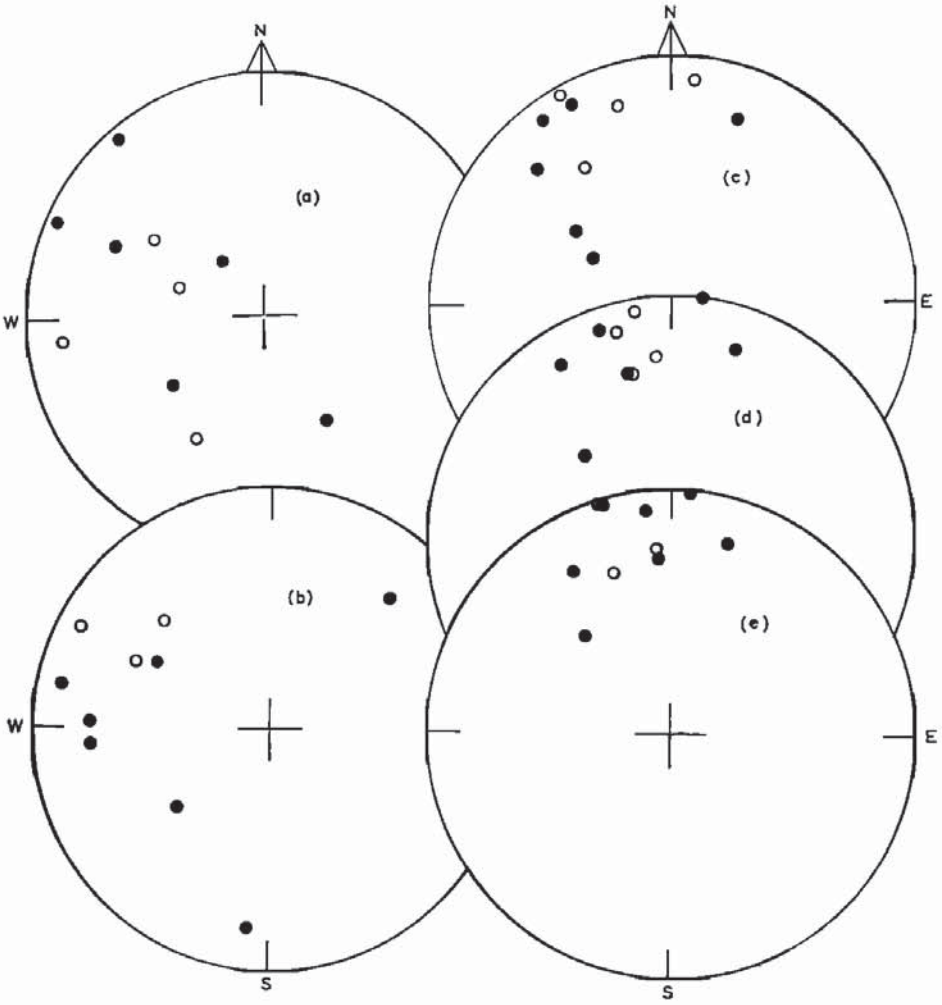


Fig. 4. Stereographic projection of directions of magnetization at site D. (*a*) NRM, after successive demagnetization in peak alternating fields of (*b*) 300, (*c*) 600, (*d*) 800, and (*e*) 1000 gauss. Closed circles, south-seeking direction on lower hemisphere; open circles, south-seeking direction on upper hemisphere.

Fisher's k [*Fisher*, 1953] increases slowly from 2.5 to 12.6, and the mean direction of magnetization is effectively the same at 800 and 1000 gauss.

Figure 5 shows the demagnetization curves for one specimen from each of the three sites. The specimen from site E (curve *a*) has reached a stable direction in a field of 100 gauss, and further demagnetization produces no change in direction while the moment falls off slowly in the manner of a stable TRM. Curve *c* in Figure 5 illustrates the behavior of a specimen from site 3. Little change in direction is observed up to a peak field of 150 gauss, and the moment has already dropped to 2.7 per cent of the NRM. From 200 to 400 gauss the direction of magnetization moves along a great circle through an angle of 90°. At this point the moment is only 0.38 per cent of its original NRM. Between peak fields of 400 and 800 gauss the direction of magnetization remains constant while the moment drops to 0.30 per cent of its NRM value. The specimen from site D, shown in curve *e* of Figure 5, shows very similar effects, except that up to 100 gauss a fairly soft component is removed while the moment drops to

TABLE 2. Mean Directions at Site D after Successive Demagnetization in the Peak Alternating Fields Indicated

Symbols are the same as for Table 1; $N = 10$.

B_{ac}	D	I	R	k
NRM	95½	−4	6.408	2.5
300	102	−12	6.311	2.5
600	153½	−9	8.604	6.5
800	167½	−3½	9.205	11.3
1000	168	−5½	9.288	12.6

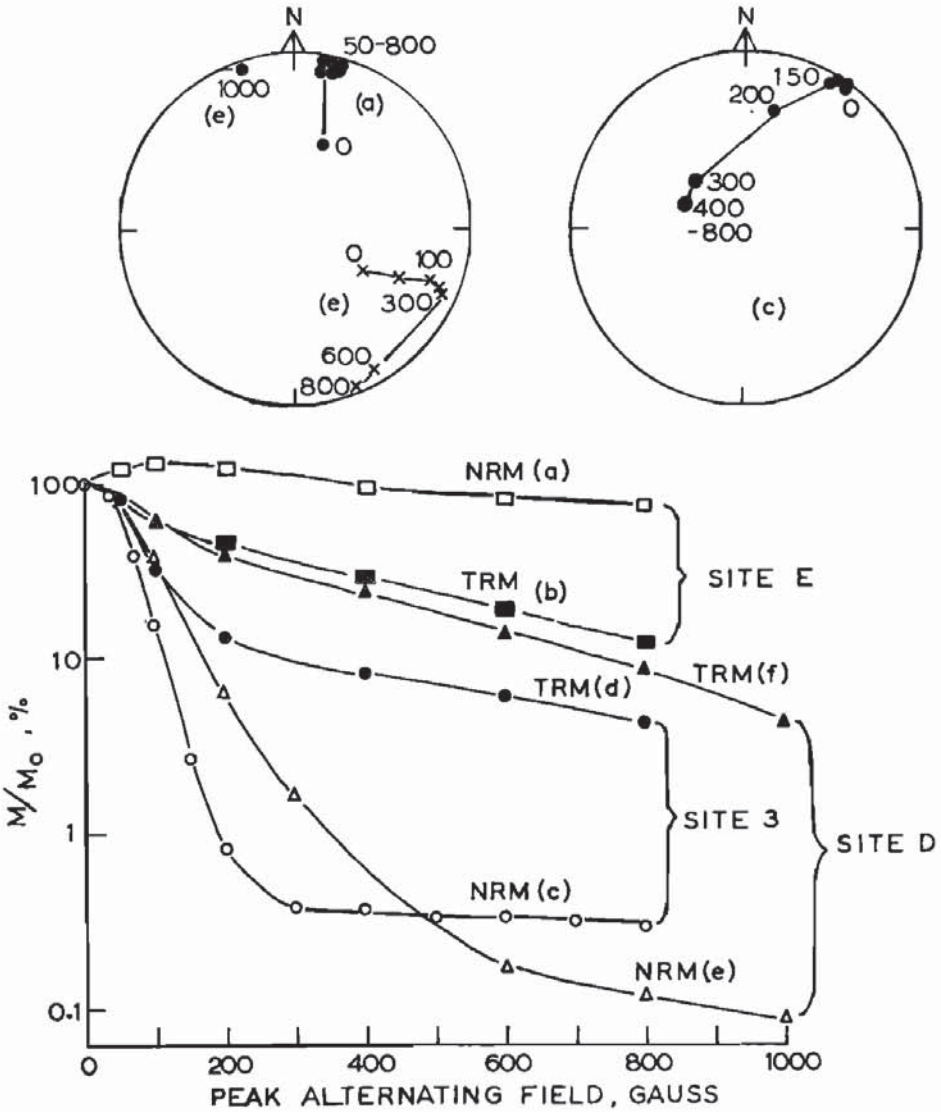


Fig. 5. Comparison of the normalized intensities of NRM and TRM in 0.32 gauss of specimens from sites 3, D, and E during alternating-field demagnetization. M/M_0 is the ratio of remaining to initial intensity after treatment in the corresponding field. Changes in direction during demagnetization of NRM are shown in lower-hemisphere stereographic projections. Solid circles represent south-seeking directions down, crosses north-seeking directions down. The peak alternating field in gauss corresponding to some of the points is also indicated.

37.5 per cent of the NRM. Between 100 and 300 gauss little change in direction is observed, but the moment drops quickly to 1.67 per cent of its NRM value. Between 300 and 1000 gauss the direction of magnetization starts to move again, but now it moves along a great circle almost at right angles to its first motion. At a peak field of 1000 gauss the moment is only 0.09 per cent of its original NRM value.

The three specimens illustrated in Figure 5 were also heated to 700°C in a nitrogen atmosphere and cooled in the earth's field. The resulting TRM was then demagnetized in increasing alternating fields, and the resultant curves are also shown in Figure 5. The shape of the demagnetization curve in each case is typical of the demagnetization of stable TRM in dolerite or basalt [*Stott*, 1963; *van Zyl et al.*, 1962]. In each case the direction of the TRM was parallel to the applied field and remained unchanged within the experimental error up to 800 or 1000 gauss. This latter observation clearly indicates the presence of high-coercivity domains.

The two demagnetization curves (*c* and *e*) in Figure 5 have the shape which typifies the

behavior of an unstable rock when subjected to alternating-field demagnetization. Clearly these two curves do not represent the demagnetization of a TRM alone. It is most probable that they represent the demagnetization of IRM due to lightning. Several factors favor this viewpoint. To obtain random directions of NRM it is necessary for different magnetizing fields to have been operating over different parts of the outcrop sampled [*Gough et al.*, 1964]. Viscous magnetization in the present earth's field would produce a planar distribution, and it is difficult to conceive of any mechanism other than lightning currents which could produce this randomizing of directions.

Lightning currents can produce magnetization of high coercivity, depending on the magnitude and proximity of the currents involved. The initial sharp drop in moment observed in Figure 5 in the two demagnetizing curves probably merely indicates that a very large component of IRM is present. This has required the use of high alternating fields, and it seems that it is possible to isolate a small minority of stable domains of high coercivity by randomizing the directions of magnetization of the other domains. By this means the use of high alternating fields has effectively removed the effect of those domains which have been reoriented by lightning currents. This often means that the residual moment is less than 1 per cent of the original NRM. Therefore it seems that the shape of the demagnetization curve has little bearing on the 'stability' of the rock specimen being subjected to alternating fields. In Figure 5, curve *e*, the change in direction observed in the specimen which moves along two great circles almost at right angles probably indicates the occurrence of two lightning currents at this site, each producing a different coercivity IRM.

In Figure 6 is shown a thermal demagnetization curve for an adjacent specimen from the same core at site D. This curve was obtained on the thermal demagnetization apparatus at Imperial College, London, described by *Wilson* [1962]. No change in direction occurs on heating the specimen up to its Curie point. This probably indicates that the blocking temperature for high coercivity domains isothermally remagnetized is either at or very near the Curie point. It seems unlikely, therefore, that thermal demagnetization will prove successful as a means of removing lightning IRM. The technique of thermal demagnetization is probably most useful where a partial thermoremanence is involved. In the present situation the only successful demagnetization technique was the use of high alternating fields.

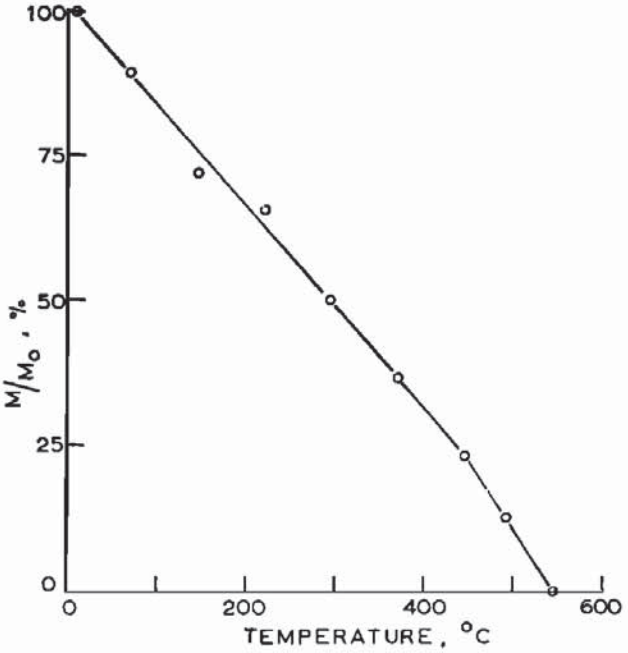


Fig. 6. Thermal demagnetization of the NRM of a specimen from site D. M/M_0 has the same meaning as in Figure 5.

At five sites it was impossible to gain any useful paleomagnetic data after applying alternating fields. In these cases it is likely that an insufficient number of high-coercivity domains were present in the specimens, so that on applying higher fields the situation was reached of the statistical resultant of the randomly oriented domain magnetizations being greater than that of the few high-coercivity domains. In these cases the lightning effects had completely remagnetized the samples.

Discussion of results. Table 1 gives the parameters at all the sites after alternating-field demagnetization. At a few sites one or two samples were excluded from the analysis. These samples had very high moments, which on demagnetization changed little up to 800 gauss either in direction or intensity. These odd samples were in contrast to the remainder at all the sites which tended to form a tight group. These few samples were probably the result of direct hits by lightning.

Figure 7 shows a lower-hemisphere stereo-

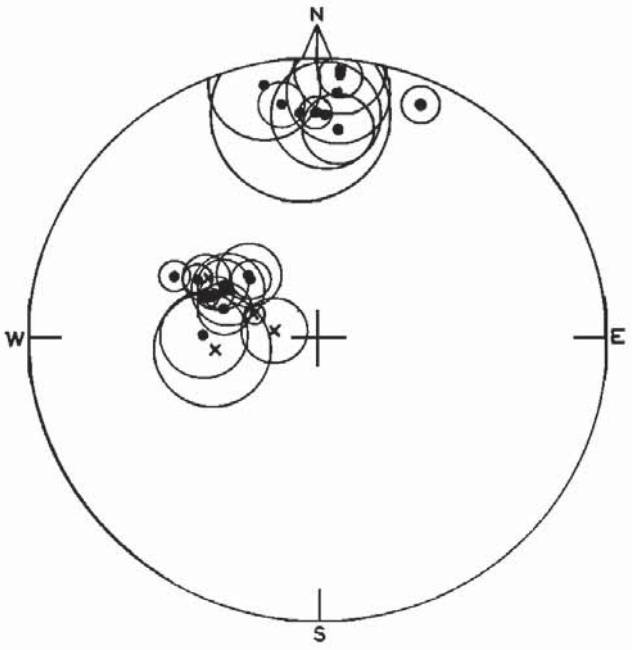


Fig. 7. Mean directions of magnetization at twenty-five sites in the Precambrian dolerites. Circles of 95 per cent confidence are plotted on a lower-hemisphere stereographic projection. Solid circles, south-seeking directions down (21 sites); crosses, north-seeking directions down (4 sites).

graphic projection of the mean directions of magnetization at the twenty-five sites listed in Table 1, together with their circles of 95 per cent confidence. The mean directions of magnetization form two well-defined groups differing by nearly 90° in direction. The set with a declination of almost 300° and an inclination of 50° are numbered 1 to 14 in Table 1 and Figure 1. This direction is characteristic of all the intrusions sampled in the basement granites and gneisses on the Rhodesian shield. These dolerites are almost entirely restricted to the Mashonaland district of Southern Rhodesia and are therefore here named the *Mashonaland dolerites*. The other directions, which are low in inclination to the south, are lettered A to J in Table 1 and Figure 1. These dolerites are always associated with and intruded into the Umkondo system and are here called the *Umkondo dolerites*.

The Great Dyke. Site 8 represents the exposure of norite from the Great Dyke in the Umsweswe River. Sites 6 and 10 from nearby dolerite sills are seen to give a similar mean direction of magnetization, which lends support to the hypothesis that the Great Dyke norite in this area has been reheated by these Mashonaland dolerites. A 200-m sill is exposed in the Umsweswe Mine immediately underlying the Umsweswe River at this point [*Worst*, 1960], but unfortunately the mine has been flooded and it was not possible to collect samples from this sill. The dolerite at site 6 forms a sheet resting on top of the Great Dyke. It is very strongly magnetic and represents the 'magnetite dolerite' of *Hess* [1950]. This sheet, as are most others in this area, is clearly not Great Dyke in age on the paleomagnetic evidence presented here, and it therefore cannot be the roof phase of the Great Dyke as Hess suggests. It is probably a coincidence that the mineral content for these dolerites falls on the mineral variation curves plotted by Hess for the Great Dyke rocks.

The Umkondo dolerites. The Umkondo dolerites have been found only in those areas in which the Umkondo system rocks are developed. The principal occurrence of this system is in the Melsetter area in the region south of Umtali (Figure 1). These dolerites again occur in the Inyanga mountains, 160 km to the north, where remnants of what have always been regarded as the northern extension of the Umkondo system are preserved. The correlation of the dolerites between these two areas confirms the supposition that the small sedimentary sequence in the Inyanga area can be correlated with the Umkondo system. In thin section these dolerites consist of pyroxene and feldspar with a well-defined ophitic relationship, and they are fairly coarse in grain. The pyroxene is dominantly colorless augite, pigeonite being rare, and the feldspar is andesine or labradorite. Subordinate amounts of quartz and apatite occur. In polished section the opaque minerals are predominantly ilmeno-magnetite. The ilmenite usually occurs exclusively as thick exsolution lamellas and granules intergrown with magnetite. At site A, however, the opaques are predominantly maghemite with ultrafine intergrowths of ilmenite.

In the Inyanga area the Umkondo dolerites form the cappings to the hills and are found to be in intimate association with the Mashonaland dolerites. Topographically higher horizons in these sills gave Umkondo directions (at sites A, B, C, and D), whereas the lower horizons (at sites 4 and 7) gave Mashonaland dolerite directions. One anomalous site (site 15) gave a

TABLE 3. Pole Positions Calculated from the Mean Directions of Magnetization
N is the number of sites, δp and δm are the semiaxes of the oval of 95 per cent confidence for the mean pole position; otherwise the symbols are the same as for Table 1.

Intrusions	N	D	I	R	k	α_{95}	Lat.	Long.	δp	δm
Mashonaland dolerites	14	292	+51	13 74	50	5½	7½°N	21°W	5	6
Umkondo dolerites	10	182½	−10½	9.843	58	6½	65½°N	39°E	3½	6½

direction which agreed with neither set of dolerites. This occurred in one of the lower horizons in the Mudzoro River on Gairezi Ranch, and it is interesting to note that the inclination observed is similar to that of the Mashonaland dolerites. It seems likely that the outcrop at this site has been rotated in a large landslide at some time after it was originally formed. The difference in declination between this site and that of the mean of the Mashonaland dolerites is 82°. This is the angle through which this outcrop may have been rotated.

The Mashonaland dolerites. These constitute the most extensively developed body of dolerite intrusions in Southern Rhodesia. Petrologically the Mashonaland dolerites are similar to the Umkondo dolerites. They consist of labradorite, hypersthene, augite, and some pigeonite. Biotite and apatite are often present as accessories. In polished section the opaque minerals are mainly ilmeno-magnetite. The ilmeno-magnetite grains have both coarse laths of granules and a grid of very fine ilmenite intergrowth. Pyrite and chalcopyrite occur rarely as small grains. Extensive sampling shows that they are found to intrude all the basement systems, the Great Dyke, and the Umkondo system. This fact could only be observed on paleomagnetic grounds. At site 12 in the Nyhodi River valley, south of Melsetter, a direction of magnetization was observed which corresponded to the Mashonaland dolerites. This site was located in fine-grained basaltic material occurring in the center of a large sill regarded by *Watson* [1964] as being part of the same sill in which sites E, F, G, and H are situated. These latter sites, as well as site I, located 1 mile away, all give Umkondo dolerite directions. Site 12 is therefore regarded as a separate intrusion into the large Umkondo sill. On normal geologic considerations, therefore, it appears that the Mashonaland dolerites were intruded later than the Umkondo dolerites. Site 12 is the only place where the Mashonaland dolerites have been found intruding both the Umkondo system and the Umkondo dolerites. It is probable that more exist.

In the Inyanga area the Mashonaland dolerites have been intruded along the same lines of weakness as the earlier Umkondo dolerites. The magma probably rose in the crust until its ascent was blocked by the earlier dolerite sheets. In the field it is impossible to distinguish between the two sets of dolerites, and future detailed mapping of this area must be based on the paleomagnetic direction of the dolerites.

The Mashonaland dolerites show both normal and reversed polarity in the ratio of 4:10. In the present instance 'normal' polarity refers to those directions having positive inclination. The mean directions of the two polarities differ by 168° ± 18°. This provides strong evidence in favor of the stability of the magnetization of these intrusions and the demagnetization techniques used in this study.

The ancient pole positions. The mean directions of magnetization for the Mashonaland dolerites and also for the Umkondo dolerites have been calculated giving each site unit weight and combining the directions irrespective of sign. These mean directions, together with the corresponding pole positions on the assumption of an axial dipole field, are given in Table 3. In the case of the Mashonaland dolerites, the sampling sites cover an area of 80,000 km², and the directions include both normal and reversed polarity. It seems likely that these sites became magnetized over an interval of time long in comparison with the secular variation, so that the mean pole position approximates the geocentric axial dipole position freed from secular variation. These dolerites intrude the Precambrian Rhodesian shield, which has suffered little tectonic movement since their intru-

sion. The geology of this part of the Rhodesian shield is known only on the basis of reconnaissance mapping, and it is difficult to determine the extent to which small tectonic movements have affected these dolerites. It seems likely, however, that very small tectonic movements of these dolerites would be randomized over the wide sampling area covered.

The Umkondo dolerites are not as widespread as the Mashonaland dolerites, but they occur over nearly 3° of latitude, and it seems likely that the ten sampling sites cover a sufficient time to average out the effects of secular variation. The dolerites intrude the Umkondo beds, which usually dip gently at about 5° in a southeasterly direction. The directions of magnetization have therefore been corrected in each case for the attitude of the adjacent sediments at each sampling site. The corrections are very small; Fisher's k for the directions changes from 57 to 58 on application of the corrections. This change is not significant, but the corrections are so small that this is to be expected. The directions cited for the various Umkondo dolerite sites in Table 1 refer to the directions after correction for the attitude of the sampling sites.

It has been pointed out by *Creer et al.* [1959] and *McElhinny and Gough* [1963] that, in the absence of sufficient evidence concerning the manner in which the geomagnetic pole moves round the geographic pole, it is reasonable to assume a symmetrical distribution of poles, as calculated from each site, rather than an oval distribution. This distribution arises from the assumption of a Fisherian distribution for the directions of magnetization. Therefore in Table 4 the pole positions for these two sets of intrusions have been calculated from the individual site pole positions, assuming that these have a Fisherian distribution.

TABLE 4. Pole Positions Calculated by Combining the Site Pole Positions

N is the number of sites, A_{95} is the semiangle of the cone of 95 per cent confidence for the mean pole position, K is the precision of site poles; otherwise the symbols are the same as for Table 1.

Intrusions	N	R	K	A_{95}	Lat.	Long.
Mashonaland dolerites	14	13.54	28	7½	7°N	20°W
Umkondo dolerites	10	9.864	66	6	65°N	42½°E

TABLE 5. Potassium-Argon Age Determinations on the Mashonaland Dolerites
Experimental errors are given.

Site No.	Age and Error, m.y.
3	1638 ± 26
9	1433 ± 22
10	1521 ± 21

The age of the Umkondo system. Dr. J. A. Miller at Cambridge has kindly made K-Ar determinations on our samples of dolerite from the Mashonaland dolerites (sites 3, 9, and 10). The results are listed in Table 5. They show that the Mashonaland dolerites are at least 1600 m.y. old. Samples of Umkondo dolerite gave inconclusive results and apparently have suffered loss of Ar. However, the paleomagnetic evidence presented in this paper shows that the Umkondo dolerites are older than the Mashonaland dolerites; consequently, the Umkondo dolerites and therefore the Umkondo system itself must be at least 1600 m.y. old. This system is usually regarded as Upper Precambrian, but it must now be placed in the Middle Precambrian. Recent correlation by *Swift* [1961] of the Umkondo system with the Transvaal system in South Africa, mainly on lithological grounds, appears to be more tenable. *Nicolaysen* [1962] regards the Transvaal system as being approximately 2000 m.y. old, that is, just older than the Bushveld igneous complex and the Great Dyke.

Acknowledgments. It is a pleasure to acknowledge the assistance of Mr. R. L. A. Watson of the Southern Rhodesia Geological Survey, who selected all the sampling sites in the Melsetter area of the Umkondo system. We wish to thank Professor A. F. Buddington of Princeton University, who very kindly looked at some polished sections, and Dr. R. L. Wilson of Imperial College, London, for the use of his thermal demagnetization apparatus. Thanks are also due Mr. H. R. Spall, who did the thermal demagnetization experiments.

This work was undertaken with the assistance of grant G.12426 from the National Science Foundation, and their financial assistance is gratefully acknowledged.

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(Manuscript received January 3, 1964.)